

SOUNIC AKKARAJU

+31 611732672 ♦ Enschede, Netherlands

imsounic.dev@gmail.com ♦ [LinkedIn](#) ♦ [Portfolio Website](#) ♦ [GitHub](#)

SUMMARY

Master's student in Computer Science (specializing in Data Science & Technology) at University of Twente, with a strong foundation in AI, Machine Learning, Deep Learning, and Big Data technologies. Experienced in backend development and full ML pipelines, with previous internship at CoreTek Labs and hands-on project work in NLP, healthcare prediction, distributed computing, and transformer-based model evaluation.

WORK EXPERIENCE

Software Engineering Intern

Jul 2023 - Sep 2023

CoreTek Labs

India

- Spearheaded the development of an AI chatbot, leveraging Django REST framework to create a seamless backend infrastructure.
- Acquired expertise in web scraping techniques and utilized the scraped data to train the chatbot, enhancing its knowledge base for the company's website.
- Worked closely with the back-end development team, fostering a collaborative environment that facilitated knowledge sharing and problem-solving.

PROJECTS

Miscarriage Prediction Using Ensemble Deep Learning Models

Dec 2024 - May 2025

Bachelor Thesis

- Developed novel NearSMOTE class-balancing technique combining SMOTE, NearMiss, and Tomek links to handle severe class imbalance in healthcare prediction tasks.
- Built ensemble deep learning system combining TabTransformer, FT-Transformer, and TabNet architectures, achieving 93.1% AUC-ROC on NFHS-5 national health survey data.
- Implemented end-to-end ML pipeline: LASSO feature selection, class balancing, model training (PyTorch on A100 GPU), and SHAP-based interpretability analysis.
- Identified actionable clinical insights through SHAP analysis — prenatal care and health checks as top protective factors against pregnancy complications.

Exploring In-Context Learning and LoRA vs BERT

Sep 2025 - Nov 2025

Natural Language Processing Project

- Collaborated on 2-person research project comparing parameter-efficient fine-tuning approaches across classification, QA, and reasoning tasks using 10 benchmark datasets.
- Implemented BERT/DistilBERT fine-tuning pipeline achieving 93.7% accuracy on AG News; analyzed cross-task transfer limitations (near-random performance on out-of-domain tasks).
- Conducted systematic in-context learning experiments on Gemma3 models (270M–4B parameters), evaluating k-shot performance (k=0,5,10,25) across multiple decoding strategies.
- Quantified efficiency tradeoffs: BERT delivers 190x higher throughput (3837 vs 20 tokens/sec) but zero cross-task transfer; ICL context expansion increases latency 26-40%.
- Co-authored research article documenting deployment recommendations for single-task vs multi-task production systems.

CleanSlate — Automated Chore Management Mobile App

Personal Project

Jun 2025 - Present

- Built a cross-platform mobile app (Flutter/Dart) with Supabase backend serving PostgreSQL database with Row-Level Security, real-time subscriptions, and Edge Functions.
- Integrated Google Calendar API with OAuth 2.0 for automatic chore syncing, handling token refresh and calendar event CRUD operations.
- Designed and implemented 7-table relational database schema supporting households, role-based permissions, recurring chores, and user preference-based scheduling.
- Architected app using feature-first structure with Repository pattern, separating UI, business logic, and data layers across 15+ feature modules.

Crypto Market Microstructure Analysis

Managing Big Data Project

Dec 2025 - Feb 2026

- Built distributed data pipeline using PySpark on university HDFS cluster to process 32GB Binance full trading history across 1000+ cryptocurrency pairs.
- Engineered minute-level features: log returns, Amihud illiquidity, Parkinson volatility, taker ratios, and zero-volume flags from raw trade-level data.
- Developed 3 analytical Spark jobs: stress vs non-stress liquidity comparison, BTC/ETH shock propagation lag analysis (0–120 min), and regime-dependent sensitivity estimation.
- Key findings: Large-cap volatility amplifies 19x during stress; volatility shocks propagate instantly (836/984 pairs at lag 0) while liquidity takes 2 hours to recover.
- Implemented efficient processing with adaptive query execution, partition strategies, and Parquet columnar storage for scalable cluster deployment.

EDUCATION

MSc Computer Science (Data Science & Technologies)

University of Twente, Netherlands

Sep 2025 - Jun 2027

- Core Modules: Machine Learning, Deep Learning, Natural Language Processing, Big Data

BSc Computer Science and Artificial Intelligence

Aberystwyth University, United Kingdom

Sep 2023 - Jun 2025

- Core Modules: Scientific Python (1st), Software Engineering (1st), Modelling Persistent Data (1st)

TECHNICAL SKILLS

Programming Languages: Python, Dart, JavaScript/TypeScript, Java, SQL, HTML/CSS

ML/AI: PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, SHAP, Pandas, NumPy, lm-eval

Frameworks: Flutter, Django REST, React, Next.js, PySpark, PostgreSQL, Supabase, Git

Cloud & Data: Google Cloud (OAuth 2.0, Calendar API), Supabase (Auth, Storage, Realtime), HDFS, Parquet

Concepts: Deep Learning, NLP, Transformers, Distributed Computing, REST APIs, Database Design

LANGUAGES

English (Fluent) • Hindi (Fluent) • Telugu (Native) • Dutch (Learning)